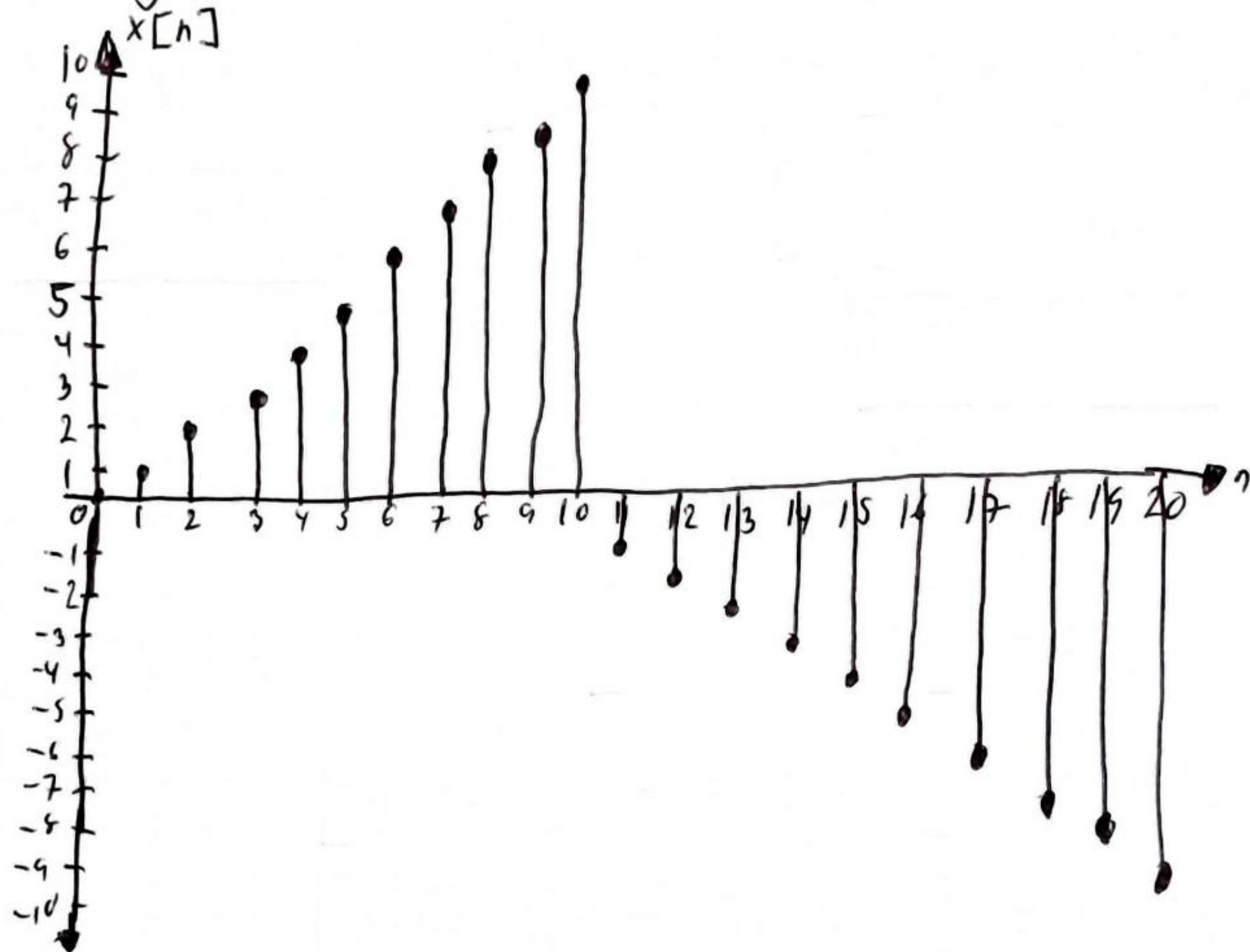


SEATWORK #1

$$\begin{aligned} 1. x[n] &= n; 0 \leq n \leq 10 \\ &= 10-n; 11 \leq n \leq 20 \\ &= 0; \text{otherwise} \end{aligned}$$

a. Draw the given signal



b. Determine the power or energy of the signal

$$E_1 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2$$

$$E_1 = 385$$

$$E_2 = (-1)^2 + (-2)^2 + (-3)^2 + (-4)^2 + (-5)^2 + (-6)^2 + (-7)^2 + (-8)^2 + (-9)^2 + (-10)^2$$

$$E_2 = 385$$

$$E_1 + E_2 = 385 + 385$$

$$\boxed{E = 770}$$

2. Obtain the odd and even components of the following signals

a. $x(t) = 20t^3$

$$E(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$E(t) = \frac{1}{2} [20t^3 + 20(-t)^3]$$

$$E(t) = \frac{1}{2} [20t^3 - 20t^3]$$

$$E(t) = \frac{1}{2} [0]$$

$$\boxed{E(t) = 0}$$

$$O(t) = \frac{1}{2} [x(t) - x(-t)]$$

$$O(t) = \frac{1}{2} [20t^3 - 20(-t)^3]$$

$$O(t) = \frac{1}{2} [20t^3 + 20t^3]$$

$$O(t) = \frac{1}{2} [40t^3]$$

$$\boxed{O(t) = 20t^3}$$

b. $x(t) = 10 \sin(20\pi t)$

$$E(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$E(t) = \frac{1}{2} [10 \sin(20\pi t) + 10 \sin(20\pi(-t))]$$

$$E(t) = \frac{1}{2} [10 \sin(20\pi t) - 10 \sin(20\pi t)]$$

$$E(t) = \frac{1}{2} [0]$$

$$\boxed{E(t) = 0}$$

$$O(t) = \frac{1}{2} [x(t) - x(-t)]$$

$$O(t) = \frac{1}{2} [10 \sin(20\pi t) - 10 \sin(-20\pi t)]$$

$$O(t) = \frac{1}{2} [10 \sin(20\pi t) + 10 \sin(20\pi t)]$$

$$O(t) = \frac{1}{2} [20 \sin(20\pi t)]$$

$$\boxed{O(t) = 10 \sin(20\pi t)}$$